

Chapter 3: Complex Numbers

EXERCISE 3.1 [PAGES 37 - 38]

Exercise 3.1 | Q 1.1 | Page 37

Write the conjugates of the following complex numbers: $3 + i$

SOLUTION

Conjugate of $(3 + i)$ is $(3 - i)$

Exercise 3.1 | Q 1.2 | Page 37

Write the conjugates of the following complex numbers: $3 - i$

Conjugate of $(3 - i)$ is $(3 + i)$

Exercise 3.1 | Q 1.3 | Page 37

Write the conjugates of the following complex numbers: $-\sqrt{5} - \sqrt{7}i$

SOLUTION

Conjugate of $(-\sqrt{5} - \sqrt{7}i)$ is $(-\sqrt{5} + \sqrt{7}i)$

Exercise 3.1 | Q 1.4 | Page 37

Write the conjugates of the following complex numbers: $-\sqrt{-5}$

SOLUTION

$$-\sqrt{-5} = -\sqrt{5} \times \sqrt{-1} = \sqrt{-5}i$$

Exercise 3.1 | Q 1.5 | Page 37

Write the conjugates of the following complex numbers: $5i$

SOLUTION

Conjugate of $5i$ is $-5i$

Exercise 3.1 | Q 1.6 | Page 37

Write the conjugates of the following complex numbers: $\sqrt{5} - i$

SOLUTION

Conjugate of $\sqrt{5} - i$ is $\sqrt{5} + i$

Exercise 3.1 | Q 1.7 | Page 37

Write the conjugates of the following complex numbers: $\sqrt{2} + \sqrt{3}i$

SOLUTION

Conjugate of $\sqrt{2} + \sqrt{3}i$ is $\sqrt{2} - \sqrt{3}i$

Exercise 3.1 | Q 2.1 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :
 $(1 + 2i)(-2 + i)$

SOLUTION

$$\begin{aligned}(1 + 2i)(-2 + i) &= -2 + i - 4i + 2i^2 \\ &= -2 - 3i + 2(-1) \quad \dots [\because i^2 = -1] \\ \therefore (1 + 2i)(-2 + i) &= -4 - 3i \\ \therefore a &= -4 \text{ and } b = -3\end{aligned}$$

Exercise 3.1 | Q 2.2 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :

$$\frac{i(4 + 3i)}{1 - i}$$

SOLUTION

$$\begin{aligned}\frac{i(4 + 3i)}{1 - i} &= \frac{4i + 3i^2}{1 - i} \\ &= \frac{-3 + 4i}{1 - i} \quad \dots [\because i^2 = -1]\end{aligned}$$

$$\begin{aligned}
&= \frac{(-3 + 4i)(1 + i)}{(1 - i)(1 + i)} \\
&= \frac{3 - 3i + 4i + 4i^2}{1 - i^2} \\
&= \frac{-3 + i + 4(-1)}{1 - (-1)} \quad \dots[\because i^2 = -1] \\
&= \frac{-7 + i}{2} \\
\therefore \frac{i(4 + 3i)}{1 - i} &= \frac{-7}{2} + \frac{1}{2}i \\
\therefore a &= \frac{-7}{2} \text{ and } b = \frac{1}{2}.
\end{aligned}$$

Exercise 3.1 | Q 2.3 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :

$$\frac{(2 + i)}{(3 - i)(1 + 2i)}$$

SOLUTION

$$\begin{aligned}
\frac{(2 + i)}{(3 - i)(1 + 2i)} &= \frac{2 + i}{3 + 6i - i - 2i^2} \\
&= \frac{2 + i}{3 + 5i - 2(-1)} \quad \dots[\because i^2 = -1] \\
&= \frac{2 + i}{5 + 5i} \\
&= \frac{2 + i}{5(1 + i)} = \frac{(2 + i)(1 - i)}{5(1 + i)(1 - i)} \\
&= \frac{2 - 2i + i - i^2}{5(1 - i^2)}
\end{aligned}$$

$$\begin{aligned}
&= \frac{2 - i - (-1)}{5[1 - (-1)]} \quad \dots[\because i^2 = -1] \\
&= \frac{3 - i}{10} \\
\therefore \frac{2 + i}{(3 - i)(1 + 2i)} &= \frac{3}{10} - \frac{1}{10}i \\
\therefore a &= \frac{3}{10} \text{ and } b = \frac{-1}{10}.
\end{aligned}$$

Exercise 3.1 | Q 2.4 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :

$$\frac{3 + 2i}{2 - 5i} + \frac{3 - 2i}{2 + 5i}$$

SOLUTION

$$\begin{aligned}
&\frac{3 + 2i}{2 - 5i} + \frac{3 - 2i}{2 + 5i} \\
&= \frac{(3 + 2i)(2 + 5i) + (2 - 5i)(3 - 2i)}{(2 - 5i)(2 + 5i)} \\
&= \frac{6 + 15i + 4i + 10i^2 + 6 - 4i - 15i + 10i^2}{4 - 25i^2} \\
&= \frac{12 + 20i^2}{4 - 25i^2} \\
&= \frac{12 + 20(-1)}{4 - 25(-1)} \quad \dots[\because i^2 = -1] \\
&= \frac{-8}{29} \\
\therefore \frac{3 + 2i}{2 - 5i} + \frac{3 - 2i}{2 + 5i} &= \frac{-8}{29} + 0i \\
\therefore a &= \frac{-8}{29} \text{ and } b = 0
\end{aligned}$$

Exercise 3.1 | Q 2.5 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :

$$\frac{2 + \sqrt{-3}}{4 + \sqrt{-3}}$$

SOLUTION

$$\begin{aligned} \frac{2 + \sqrt{-3}}{4 + \sqrt{-3}} &= \frac{2 + \sqrt{3}i}{4 + \sqrt{3}i} \\ &= \frac{(2 + \sqrt{3}i)(4 - \sqrt{3}i)}{(4 + \sqrt{3}i)(4 - \sqrt{3}i)} \\ &= \frac{8 - 2\sqrt{3}i + 4\sqrt{3}i - 3i^2}{16 - 3i^2} \\ &= \frac{8 + 2\sqrt{3}i - 3(-1)}{16 - 3(-1)} \quad \dots[\because i^2 = -1] \\ &= \frac{11 + 2\sqrt{3}i}{19} \\ \therefore \frac{2 + \sqrt{-3}}{4 + \sqrt{-3}} &= \frac{11}{19} + \frac{2\sqrt{3}}{19}i \\ \therefore a &= \frac{11}{19} \text{ and } b = \frac{2\sqrt{3}}{19} \end{aligned}$$

Exercise 3.1 | Q 2.6 | Page 37

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :
 $(2 + 3i)(2 - 3i)$

SOLUTION

$$\begin{aligned} (2 + 3i)(2 - 3i) &= 4 - 9i^2 \\ &= 4 - 9(-1) \quad \dots[\because i^2 = -1] \\ &= 4 + 9 = 13 \\ \therefore (2 + 3i)(2 - 3i) &= 13 + 0i \\ \therefore a &= 13 \text{ and } b = 0 \end{aligned}$$

Exercise 3.1 | Q 2.7 | Page 38

Express the following in the form of $a + ib$, $a, b \in \mathbb{R}$, $i = \sqrt{-1}$. State the values of a and b :

$$\frac{4i^8 - 3i + 3}{3i^{11} - 4i^{10} - 2}$$

SOLUTION

$$\frac{4i^8 - 3i + 3}{3i^{11} - 4i^{10} - 2} = \frac{4(i^4)^2 - 3(i^4)^2 \cdot i + 3}{3(i^4)^2 \cdot i^3 - 4(i^4)^2 \cdot i^2 - 2}$$

Since, $i^2 = -1$, $i^3 = -i$ and $i^4 = 1$

$$\therefore \frac{4i^8 - 3i^9 + 3}{3i^{11} - 4i^{10} - 2} = \frac{4(1)^2 - 3(1)^2 \cdot i + 3}{3(1)^2(-i) - 4(1)^2(-1) - 2}$$

$$= \frac{4 - 3i + 3}{-3i + 4 - 2}$$

$$= \frac{7 - 3i}{2 - 3i}$$

$$= \frac{(7 - 3i)(2 - 3i)}{(2 - 3i)(2 + 3i)}$$

$$= \frac{14 + 21i - 6i - 9i^2}{4 - 9(-1)}$$

$$= \frac{14 + 15i - 9(-1)}{4 - 9(-1)}$$

$$= \frac{23 + 15i}{13}$$

$$\therefore \frac{4i^8 - 3i^9 + 3}{3i^{11} - 4i^{10} - 2} = \frac{23}{13} + \frac{15}{13}i$$

$$\therefore a = \frac{23}{13} \text{ and } b = \frac{15}{13}$$

Exercise 3.1 | Q 3 | Page 38

Show that $(-1 + \sqrt{3}i)^3$ is a real number.

SOLUTION

$$\begin{aligned} & (-1 + \sqrt{3}i)^3 \\ &= (-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3 \quad \dots [(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3] \\ &= -1 + 3\sqrt{3}i - 3(3i^2) + 3\sqrt{3}i^3 \\ &= -1 + 3\sqrt{3}i - 3(3i^2) + 3\sqrt{3}i \quad \dots [\because i^2 = -1, i^3 = -i] \\ &= -1 + 9 \\ &= 8, \text{ which is a real number.} \end{aligned}$$

Exercise 3.1 | Q 4.1 | Page 38

Evaluate the following: i^{35}

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 $i^{35} = (i^4)^8 (i^2)i = (1)^8 (-1)i = -i$

Exercise 3.1 | Q 4.2 | Page 38

Evaluate the following: i^{888}

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 $i^{888} = (i^4)^{222} = (1)^{222} = 1$

Exercise 3.1 | Q 4.3 | Page 38

Evaluate the following: i^{93}

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 $i^{93} = (i^4)^{23} \cdot i = (1)^{23} \cdot i = i$

Exercise 3.1 | Q 4.4 | Page 38

Evaluate the following: i^{116}

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 $i^{116} = (i^4)^{29} = (1)^{29} = 1$

Exercise 3.1 | Q 4.5 | Page 38

Evaluate the following: i^{403}

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
 $i^{403} = (i^4)^{100} (i^2)i = (1)^{100} (-1)i = -i$

Exercise 3.1 | Q 4.6 | Page 38

Evaluate the following: $\frac{1}{i^{58}}$

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$

$$\frac{1}{i^{58}} = \frac{1}{(i^4)^{14} \cdot i^2} = \frac{1}{(1)^{14}(-1)} = -1$$

Exercise 3.1 | Q 4.7 | Page 38

Evaluate the following: $i^{30} + i^{40} + i^{50} + i^{60}$

SOLUTION

We know that, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$

$$\begin{aligned} &= (i^4)^7 i^2 + (i^4)^{10} + (i^4)^{12} i^2 + (i^4)^{15} \\ &= (1)^7 (-1) + (1) + (1)^{10} + (1)^{12} (-1) + (1)^{15} \\ &= -1 + 1 - 1 + 1 \\ &= 0. \end{aligned}$$

Exercise 3.1 | Q 5 | Page 38

Show that $1 + i^{10} + i^{20} + i^{30}$ is a real number.

SOLUTION

$$\begin{aligned} &1 + i^{10} + i^{20} + i^{30} \\ &= 1 + (i^4)^2 \cdot i^2 + (i^4)^5 + (i^4)^7 \cdot i^2 \\ &= 1 + (1)^2 (-1) + (1)^5 + (1)^7 (-1) \quad \dots [\because i^4 = 1, i^2 = -1] \\ &= 1 - 1 + 1 - 1 \\ &= 0, \text{ which is a real number.} \end{aligned}$$
Exercise 3.1 | Q 6.1 | Page 38

Find the value of $i^{49} + i^{68} + i^{89} + i^{110}$

SOLUTION

$$\begin{aligned} &i^{49} + i^{68} + i^{89} + i^{110} \\ &= (i^4)^{12} \cdot i + (i^4)^{17} + (i^4)^{22} \cdot i + (i^4)^{27} \cdot i^2 \\ &= (1)^{12} \cdot i + (1)^{17} + (1)^{22} \cdot i + (1)^{27} (-1) \quad \dots [\because i^4 = 1, i^2 = -1] \end{aligned}$$

$$= i + 1 + i - 1$$

$$= 2i$$

Exercise 3.1 | Q 6.2 | Page 38

Find the value of $i + i^2 + i^3 + i^4$

SOLUTION

$$i + i^2 + i^3 + i^4$$

$$= i + i^2 + i^2 \cdot i + i^4$$

$$= i - 1 - i + 1 \quad \dots [\because i^2 = -1, i^4 = 1]$$

$$= 0.$$

Exercise 3.1 | Q 7 | Page 38

Find the value of $1 + i^2 + i^4 + i^6 + i^8 + \dots + i^{20}$.

SOLUTION

$$1 + i^2 + i^4 + i^6 + i^8 + \dots + i^{20}$$

$$= 1 + (i^2 + i^4) + (i^6 + i^8) + (i^{10} + i^{12}) + (i^{14} + i^{16}) + (i^{18} + i^{20})$$

$$= 1 + [i^2 + (i^2)^2] + [(i^2)^3 + (i^2)^4] + [(i^2)^5 + (i^2)^6] + [(i^2)^7 + (i^2)^8] + [(i^2)^9 + (i^2)^{10}]$$

$$= 1 + [-1 + (-1)^2] + [(-1)^3 + (-1)^4] + [(-1)^5 + (-1)^6] + [(-1)^7 + (-1)^8] + [(-1)^9 + (-1)^{10}] \quad \dots [\because i^2 = -1]$$

$$= 1 + (-1 + 1) + (-1 + 1) + (-1 + 1) + (-1 + 1) + (-1 + 1)$$

$$= 1 + 0 + 0 + 0 + 0 + 0$$

$$= 1.$$

Exercise 3.1 | Q 8.1 | Page 38

Find the values of x and y which satisfy the following equations ($x, y \in \mathbb{R}$): $(x + 2y) + (2x - 3y)i + 4i = 5$

SOLUTION

$$(x + 2y) + (2x - 3y)i + 4i = 5$$

$$\therefore (x + 2y) + (2x - 3y)i = 5 - 4i$$

Equating real and imaginary parts, we get

$$x + 2y = 5 \quad \dots (i)$$

and $2x - 3y = -4 \quad \dots (ii)$

Equation (i) $\times 2$ - equation (ii) gives

$$7y = 14$$

$$\therefore y = 2$$

Putting $y = 2$ in (i), we get

$$x + 2(2) = 5$$

$$\therefore x + 4 = 5$$

$$\therefore x = 1$$

$$\therefore x = 1 \text{ and } y = 2.$$

Exercise 3.1 | Q 8.2 | Page 38

Find the values of x and y which satisfy the following equations ($x, y \in \mathbb{R}$):

$$\frac{x+1}{1+i} + \frac{y-1}{1-i} = i$$

SOLUTION

$$\frac{x+1}{1+i} + \frac{y-1}{1-i} = i$$

$$\therefore \frac{(x+1)(1-i) + (y-1)(1+i)}{(1+i)(1-i)} = i$$

$$\therefore \frac{x - xi + 1 - i + y + yi - 1 - i}{1 - i^2}$$

$$\therefore \frac{(x+y) + (y-x-2)i}{1 - (-1)} = i \quad \dots [\because i^2 = -1]$$

$$\therefore (x+y) + (y-x-2)i = 2i$$

$$\therefore (x+y) + (y-x-2)i = 0 + 2i$$

Equating real and imaginary parts, we get

$$x + y = 0 \text{ and } y - x - 2 = 2$$

$$\therefore x + y = 0 \quad \dots (i)$$

$$\text{and } -x + y = 4 \quad \dots (ii)$$

Adding (i) and (ii), we get

$$2y = 4$$

$$\therefore y = 2$$

Putting $y = 2$ in (i), we get

$$x + 2 = 0$$

$$\therefore x = -2$$

$$\therefore x = -2 \text{ and } y = 2.$$

Exercise 3.1 | Q 9.1 | Page 38

Find the value of: $x^3 - x^2 + x + 46$, if $x = 2 + 3i$

SOLUTION

$$x = 2 + 3i$$

$$\therefore x - 2 = 3i$$

$$\therefore (x-2)^2 = 9i^2$$

$$\therefore x^2 - 4x + 4 = 9(-1) \quad \dots [\because i^2 = -1]$$

$$\therefore x^2 - 4x + 13 = 0 \quad \dots (i)$$

$$\begin{array}{r}
 \overline{x+3} \\
 x^2-4x+13 \overline{) x^3+x^2+x+46} \\
 \underline{x^3-4x^2+13x} \\
 3x^2-12x+46 \\
 \underline{3x^2-12x+39} \\
 7
 \end{array}$$

$$\begin{aligned}
 &\therefore x^3 - x^2 + x + 46 \\
 &= (x^2 - 4x + 13)(x + 3) + 7 \\
 &= 0(x + 3) + 7 \quad \dots[\text{From (i)}] \\
 &= 7.
 \end{aligned}$$

Exercise 3.1 | Q 9.2 | Page 38

Find the value of: $2x^3 - 11x^2 + 44x + 27$, if $x = \frac{25}{3 - 4i}$

SOLUTION

$$\begin{aligned}
 x &= \frac{25}{3 - 4i} \\
 \therefore x &= \frac{25(3 + 4i)}{(3 - 4i)(3 + 4i)} \\
 &= \frac{25(3 + 4i)}{9 - 16i^2} \\
 &= \frac{25(3 + 4i)}{9 - 16(-1)} \quad \dots[\because i^2 = -1] \\
 &= \frac{25(3 + 4i)}{25}
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= 3 + 4i \\
 \therefore x - 3 &= 4i \\
 \therefore (x - 3)^2 &= 16i^2
 \end{aligned}$$

$$\begin{aligned}\therefore x^2 - 6x + 9 &= 16(-1) \quad \dots [\because i^2 = -1] \\ \therefore x^2 - 6x + 25 &= 0\end{aligned}$$

$$\begin{array}{r}
 2x + 1 \\
 x^2 - 6x + 25 \overline{) 2x^3 - 11x^2 + 44x + 27} \\
 \underline{2x^3 - 12x^2 + 50x} \\
 x^2 - 6x + 27 \\
 \underline{x^2 - 6x + 25} \\
 2
 \end{array}$$

EXERCISE 3.2 [PAGE 40]

Exercise 3.2 | Q 1.1 | Page 40

Find the square root of the following complex numbers: $-8 - 6i$

SOLUTION

Let $\sqrt{-8 - 6i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$-8 - 6i = (a + bi)^2$$

$$\therefore -8 - 6i = a^2 + b^2i^2 + 2abi$$

$$\therefore -8 - 6i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = -8 \text{ and } 2ab = -6$$

$$\therefore a^2 - b^2 = -8 \text{ and } b = \frac{-3}{a}$$

$$\therefore a^2 - \left(-\frac{3}{a}\right)^2 = -8$$

$$\therefore a^2 - \frac{9}{a^2} = -8$$

$$\therefore a^4 - 9 = -8a^2$$

$$\therefore a^4 + 8a^2 - 9 = 0$$

$$\therefore (a^2 + 9)(a^2 - 1) = 0$$

$$\therefore a^2 = -9 \text{ or } a^2 = 1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -9$$

$$\therefore a^2 = 1$$

$$\therefore a = \pm 1$$

$$\text{when } a = 1, b = \frac{-3}{1} = -3$$

$$\text{when } a = -1, b = \frac{-3}{-1} = 3$$

$$\therefore \sqrt{-8 - 6i} = \pm (1 - 3i).$$

Exercise 3.2 | Q 1.2 | Page 40

Find the square root of the following complex numbers: $7 + 24i$

SOLUTION

Let $\sqrt{7 + 24i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$7 + 24i = (a + bi)^2$$

$$\therefore 7 + 24i = a^2 + b^2i^2 + 2abi$$

$$\therefore 7 + 24i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 7 \text{ and } 2ab = 24$$

$$\therefore a^2 - b^2 = 7 \text{ and } b = \frac{12}{a}$$

$$\therefore a^2 - \left(\frac{12}{a}\right)^2 = 7$$

$$\therefore a^2 - \frac{144}{a^2} = 7$$

$$\therefore a^4 - 144 = 7a^2$$

$$\therefore a^4 - 7a^2 - 144 = 0$$

$$\therefore (a^2 - 16)(a^2 + 9) = 0$$

$$\therefore a^2 = 16 \text{ or } a^2 = -9$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -9$$

$$\therefore a^2 = 16$$

$$\therefore a = \pm 4$$

$$\text{When } a = 4, b = \frac{12}{4} = 3$$

$$\text{When } a = -4, b = \frac{12}{-4} = -3$$

$$\therefore \sqrt{7 + 24i} = \pm (4 + 3i).$$

Exercise 3.2 | Q 1.3 | Page 40

Find the square root of the following complex numbers: $1 + 4\sqrt{3}i$

SOLUTION

Let $\sqrt{1 + 4\sqrt{3}i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$1 + 4\sqrt{3}i = (a + bi)^2$$

$$\therefore 1 + 4\sqrt{3}i = a^2 + b^2i^2 + 2abi$$

$$\therefore 1 + 4\sqrt{3}i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 1 \text{ and } 2ab = 4\sqrt{3}$$

$$\therefore a^2 - b^2 = 1 \text{ and } b = \frac{2\sqrt{3}}{a}$$

$$\therefore a^2 - \left(\frac{2\sqrt{3}}{a}\right)^2 = 1$$

$$\therefore a^2 - \frac{12}{a^2} = 1$$

$$\therefore a^4 - 12 = a^2$$

$$\therefore a^4 - a^2 - 12 = 0$$

$$\therefore (a^2 - 4)(a^2 + 3) = 0$$

$$\therefore a^2 = 4 \text{ or } a^2 = -3$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -3$$

$$\therefore a^2 = 4$$

$$\therefore a = \pm 2$$

$$\text{When } a = 2, b = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

$$\text{When } a = -2, b = \frac{2\sqrt{3}}{-2} = -\sqrt{3}$$

$$\therefore \sqrt{1 + 4\sqrt{3}i} = \pm(2 + \sqrt{3}i)$$

Exercise 3.2 | Q 1.4 | Page 40

Find the square root of the following complex numbers: $3 + 2\sqrt{10}i$

SOLUTION

Let $\sqrt{3 + 2\sqrt{10}i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$3 + 2\sqrt{10}i = (a + bi)^2$$

$$\therefore 3 + 2\sqrt{10}i = a^2 + b^2i^2 + 2abi$$

$$3 + 2\sqrt{10}i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 3 \text{ and } 2ab = 2\sqrt{10}$$

$$\therefore a^2 - b^2 = 3 \text{ and } b = \frac{\sqrt{10}}{a}$$

$$\therefore a^2 - \left(\frac{\sqrt{10}}{a}\right)^2 = 3$$

$$\therefore a^2 - \frac{10}{a^2} = 3$$

$$\therefore a^4 - 10 = 3a^2$$

$$\therefore a^4 - 3a^2 - 10 = 0$$

$$\therefore (a^2 - 3a^2 - 10 = 0$$

$$\therefore a^2 = 5 \text{ or } a^2 = -2$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -2$$

$$\therefore a^2 = 5$$

$$\therefore a = \pm \sqrt{5}$$

$$\text{When } a = \sqrt{5}, b = \frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2}$$

When $a = -\sqrt{5}$, $b = \frac{\sqrt{10}}{-\sqrt{5}} = -\sqrt{2}$

$$\therefore \sqrt{3 + 2\sqrt{10}i} = \pm(\sqrt{5} + \sqrt{2}i)$$

Exercise 3.2 | Q 1.5 | Page 40

Find the square root of the following complex numbers: $2(1 - \sqrt{3}i)$

SOLUTION

Let $\sqrt{2(1 - \sqrt{3}i)} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$2(1 - \sqrt{3}i) = (a + bi)^2$$

$$\therefore 2(1 - \sqrt{3}i) = a^2 + b^2i^2 + 2abi$$

$$\therefore 2 - 2\sqrt{3}i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 2 \text{ and } 2ab = -2\sqrt{3}$$

$$\therefore a^2 - b^2 = 2 \text{ and } b = -\frac{\sqrt{3}}{a}$$

$$\therefore a^2 - \frac{3}{a^2} = 2$$

$$\therefore a^4 - 3 = 2a^2$$

$$\therefore a^4 - 2a^2 - 3 = 0$$

$$\therefore (a^2 - 3)(a^2 + 1) = 0$$

$$\therefore a^2 = 3 \text{ or } a^2 = -1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -1$$

$$\therefore a^2 = 3$$

$$\therefore a = \pm\sqrt{3}$$

$$\text{When } a = \sqrt{3}, b = \frac{-\sqrt{3}}{\sqrt{3}} = -1$$

$$\text{When } a = \sqrt{3}, b = \frac{-\sqrt{3}}{-\sqrt{3}} = 1$$

$$\therefore \sqrt{2(1 - \sqrt{3}i)} = \pm(\sqrt{3} - i).$$

Exercise 3.2 | Q 2.1 | Page 40

Solve the following quadratic equation: $8x^2 + 2x + 1 = 0$

SOLUTION

Given equation is $8x^2 + 2x + 1 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 8$, $b = 2$, $c = 1$

Discriminant $= b^2 - 4ac$

$$= (2)^2 - 4 \times 8 \times 1$$

$$= 4 - 32$$

$$= -28 < 0$$

So, the given equation has complex roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{-28}}{2(8)}$$

$$= \frac{-2 \pm 2\sqrt{7}i}{16}$$

$$\therefore x = \frac{-1 \pm \sqrt{7}i}{8}$$

∴ the roots of the given equation are

$$\frac{-1 + \sqrt{7}i}{8} \text{ and } \frac{-1 - \sqrt{7}i}{8}.$$

Exercise 3.2 | Q 2.2 | Page 40

Solve the following quadratic equation: $2x^2 - \sqrt{3}x + 1 = 0$

SOLUTION

Given equation is $2x^2 - \sqrt{3}x + 1 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2, b = -\sqrt{3}, c = 1$$

$$\text{Discriminant} = b^2 - 4ac$$

$$= (-\sqrt{3})^2 - 4 \times 2 \times 1$$

$$= 3 - 8$$

$$= -5 < 0$$

So, the given equation has complex roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-\sqrt{3}) \pm \sqrt{-5}}{2 \times 2}$$

$$\therefore x = \frac{\sqrt{3} \pm \sqrt{5}i}{4}$$

∴ the roots of the given equation are

$$\frac{\sqrt{3} + \sqrt{5}i}{4} \text{ and } \frac{\sqrt{3} - \sqrt{5}i}{4}.$$

Exercise 3.2 | Q 2.3 | Page 40

Solve the following quadratic equation: $3x^2 - 7x + 5 = 0$

SOLUTION

Given equation is $3x^2 - 7x + 5 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 3, b = -7, c = 5$

Discriminant $= b^2 - 4ac$

$$= (-7)^2 - 4 \times 3 \times 5$$

$$= 49 - 60$$

$$= -11 < 0$$

So, the given equation has complex roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-7) \pm \sqrt{-11}}{2 \times 3}$$

$$\therefore x = \frac{7 \pm \sqrt{11}i}{6}$$

$$\therefore \text{the roots of the given equation are } \frac{7 \pm \sqrt{11}i}{6} \text{ and } \frac{7 - \sqrt{11}i}{6}.$$

Exercise 3.2 | Q 2.4 | Page 40

Solve the following quadratic equation: $x^2 - 4x + 13 = 0$

SOLUTION

Given equation is $x^2 - 4x + 13 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 1, b = -4, c = 13$

Discriminant $= b^2 - 4ac$

$$= (-4)^2 - 4 \times 1 \times 13$$

$$= 16 - 52$$

$$= -36 < 0$$

So, the given equation has complex roots.

These roots are given by

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{- - 4 \pm \sqrt{-36}}{2 \cdot 1} \\
 &= \frac{4 \pm 6i}{2} \\
 &= 2 \pm 3i
 \end{aligned}$$

∴ the roots of the given equation are $2 \pm 3i$ and $2 - 3i$.

Exercise 3.2 | Q 3.1 | Page 40

Solve the following quadratic equation: $x^2 + 3ix + 10 = 0$

SOLUTION

Given equation is $x^2 + 3ix + 10 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 1$, $b = 3i$, $c = 10$

Discriminant $= b^2 - 4ac$

$$= (3i)^2 - 4 \times 1 \times 10$$

$$= 9i^2 - 40$$

$$= -9 - 40 \quad \dots [\because i^2 = -1]$$

$$= -49 < 0$$

So, the given equation has complex roots.

These roots are given by

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{3i \pm \sqrt{-49}}{2(1)} \\
 \therefore x &= \frac{-3i + 7i}{2} \\
 \therefore x &= \frac{-3i + 7i}{2} \quad \text{or} \quad x = \frac{-3i - 7i}{2}
 \end{aligned}$$

$$\therefore x = 2i \text{ or } x = -5i$$

∴ the roots of the given equation are $2i$ and $-5i$.

Exercise 3.2 | Q 3.2 | Page 40

Solve the following quadratic equation: $2x^2 + 3ix + 2 = 0$

SOLUTION

Given equation is $2x^2 + 3ix + 2 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 2$, $b = 3i$, $c = 2$

Discriminant $= b^2 - 4ac$

$$= (3i)^2 - 4 \times 2 \times 2$$

$$= 9i^2 - 16$$

$$= -9 - 16 \quad \dots [\because i^2 = -1]$$

$$= -25 < 0$$

So, the given equation has complex roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3i \pm \sqrt{-25}}{2(2)}$$

$$\therefore x = \frac{-3i \pm 5i}{4}$$

$$\therefore x = \frac{-3i + 5i}{4} \quad \text{or} \quad x = \frac{-3i - 5i}{4}$$

$$\therefore x = \frac{1}{2}i \quad \text{or} \quad x = -2i$$

$\therefore$ the roots of the given equation are $\frac{1}{2}i$ and $-2i$.

Exercise 3.2 | Q 3.3 | Page 40

Solve the following quadratic equation: $x^2 + 4ix - 4 = 0$

SOLUTION

Given equation is $x^2 + 4ix - 4 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 1$, $b = 4i$, $c = -4$

Discriminant $= b^2 - 4ac$

$$= (4i)^2 - 4 \times 1 \times -4$$

$$= 16i^2 + 16$$

$$= -16 + 16 \quad \dots[\because i^2 = -1]$$

$$= 0$$

So, the given equation has equal roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4i + \sqrt{0}}{2(1)}$$

$$= \frac{-4i}{2}$$

$$\therefore x = -2i$$

$\therefore$ the roots of the given equation are $-2i$ and $-2i$.

Exercise 3.2 | Q 3.4 | Page 40

Solve the following quadratic equation: $ix^2 - 4x - 4i = 0$

SOLUTION

$$ix^2 - 4x - 4i = 0$$

Multiplying throughout by i , we get

$$i^2x^2 - 4ix - 4i^2 = 0$$

$$\therefore -x^2 - 4ix + 4 = 0 \quad \dots[\because i^2 = -1]$$

$$\therefore x^2 + 4ix - 4 = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 1, b = 4i, c = -4$$

$$\text{Discriminant} = b^2 - 4ac$$

$$= (4i)^2 - 4 \times 1 \times -4$$

$$= 16i^2 + 16$$

$$= -16 + 16 \quad \dots[\because i^2 = -1]$$

$$= 0$$

So, the given equation has equal roots.

These roots are given by

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{-4i \pm \sqrt{0}}{2(1)} \\
 &= \frac{-4i}{2}
 \end{aligned}$$

$$\therefore x = -2i$$

$\therefore$ the roots of the given equation are $-2i$ and $-2i$.

Exercise 3.2 | Q 4.1 | Page 40

Solve the following quadratic equation: $x^2 - (2 + i)x - (1 - 7i) = 0$

SOLUTION

Given equation is $x^2 - (2 + i)x - (1 - 7i) = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 1$, $b = -(2 + i)$, $c = -(1 - 7i)$

Discriminant $= b^2 - 4ac$

$$= [-(2 + i)]^2 - 4 \times 1 - (1 - 7i)$$

$$= 4 + 4i + i^2 + 4 - 28i$$

$$= 4 + 4i - 1 + 4 - 28i \quad \dots [\because i^2 = -1]$$

$$= 7 - 24i$$

So, the given equation has complex roots.

These roots are given by

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{-[-(2 + i)] \pm \sqrt{7 - 24i}}{2(1)} \\
 &= \frac{(2 + i) \pm \sqrt{7 - 24i}}{2}
 \end{aligned}$$

Let $\sqrt{7 - 24i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$7 - 24i = a^2 + i^2b^2 + 2abi$$

$$\therefore 7 - 24i = (a^2 - b^2) + 2abi \quad \dots [\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 7 \text{ and } 2ab = -24$$

$$\therefore a^2 - b^2 = 7 \text{ and } b = \frac{-12}{a}$$

$$\therefore a^2 - \left(\frac{-12}{a}\right)^2 = 7$$

$$\therefore a^2 - \frac{144}{a^2} = 7$$

$$\therefore a^4 - 144 = 7a^2$$

$$\therefore a^4 - 7a^2 - 144 = 0$$

$$\therefore (a^2 - 16)(a^2 + 9) = 0$$

$$\therefore a^2 = 16 \text{ or } a^2 = -9$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -9$$

$$\therefore a^2 = 16$$

$$\therefore a = \pm 4$$

$$\text{When } a = 4, b = \frac{-12}{4} = -3$$

$$\text{When } a = -4, b = \frac{-12}{-4} = 3$$

$$\therefore \sqrt{7 - 24i} = \pm (4 - 3i)$$

$$\therefore x = \frac{(2 + i) \pm (4 - 3i)}{2}$$

$$\therefore x = \frac{(2 + i) \pm (4 - 3i)}{2} \quad \text{or} \quad x = \frac{(2 + i) - (4 - 3i)}{2}$$

$$\therefore x = 3 - i \text{ or } x = -1 + 2i.$$

Exercise 3.2 | Q 4.2 | Page 40

Solve the following quadratic equation: $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

SOLUTION

Given equation is $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 1, b = -(3\sqrt{2} + 2i), c = 6\sqrt{2}i$$

$$\text{Discriminant} = b^2 - 4ac$$

$$= \left[-(3\sqrt{2} + 2i) \right]^2 - 4 \times 1 \times 6\sqrt{2}i$$

$$= 18 + 12\sqrt{2}i + 4i^2 - 24\sqrt{2}i$$

$$= 18 - 12\sqrt{2}i - 4 \quad \dots[\because i^2 = -1]$$

$$= 14 - 12\sqrt{2}i$$

So, the given equation has complex roots.

These roots are given by

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-[-(3\sqrt{2} + 2i)] \pm \sqrt{14 - 12\sqrt{2}i}}{2(1)} \\ &= \frac{(3\sqrt{2} + 2i) \pm \sqrt{14 - 12\sqrt{2}i}}{2} \end{aligned}$$

$$\text{Let } \sqrt{14 - 12\sqrt{2}i} = a + bi, \text{ where } a, b \in \mathbb{R}$$

Squaring on both sides, we get

$$14 - 12\sqrt{2}i = a^2 + i^2b^2 + 2abi$$

$$\therefore 14 - 12\sqrt{2}i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 14 \text{ and } 2ab = -12\sqrt{2}$$

$$\therefore a^2 - b^2 = 14 \text{ and } b = \frac{-6\sqrt{2}}{a}$$

$$\therefore a^2 - \left(\frac{-6\sqrt{2}}{a} \right)^2 = 14$$

$$\therefore a^2 - \frac{72}{a^2} = 14$$

$$\therefore a^4 - 72 = 14a^2$$

$$\therefore a^4 - 14a^2 - 72 = 0$$

$$\therefore (a^2 - 18)(a^2 + 4) = 0$$

$$\therefore a^2 = 18 \text{ or } a^2 = -4$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -4$$

$$\therefore a^2 = 18$$

$$\therefore a = \pm 3\sqrt{2}$$

$$\text{When } a = 3\sqrt{2}, b = \frac{-6\sqrt{2}}{3\sqrt{2}} = -2$$

$$\text{When } a = -3\sqrt{2}, b = \frac{-6\sqrt{2}}{-3\sqrt{2}} = 2$$

$$\therefore \sqrt{14 - 12\sqrt{2}i} = \pm (3\sqrt{2} - 2i)$$

$$\therefore x = \frac{(3\sqrt{2} + 2i) \pm (3\sqrt{2} - 2i)}{2}$$

$$\therefore x = \frac{(3\sqrt{2} + 2i) + (3\sqrt{2} - 2i)}{2}$$

$$\text{or } x = \frac{(3\sqrt{2} + 2i) - (3\sqrt{2} - 2i)}{2}$$

$$\therefore x = 3\sqrt{2} \text{ or } x = 2i.$$

Exercise 3.2 | Q 4.3 | Page 40

Solve the following quadratic equation: $x^2 - (5 - 1)x + (18 + i) = 0$

SOLUTION

Given equation is $x^2 - (5 - 1)x + (18 + i) = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 1$, $b = -(5 - i)$, $c = 18 + i$

Discriminant $= b^2 - 4ac$

$$= [-(5 - i)]^2 - 4 \times 1 \times (18 + i)$$

$$= 25 - 10i + i^2 - 72 - 4i$$

$$= 25 - 10i - 1 - 72 - 4i \quad \dots [\because i^2 = -1]$$

$$= -48 - 14i$$

So, the given equation has complex roots.

These roots are given by

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-[-(5 - i)] \pm \sqrt{-48 - 14i}}{2(1)} \\ &= \frac{(5 - i) \pm \sqrt{-48 - 14i}}{2} \end{aligned}$$

Let $\sqrt{-48 - 14i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$-48 - 14i = a^2 + b^2i^2 + 2abi$$

$$\therefore -48 - 14i = (a^2 - b^2) + 2abi \quad \dots [\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = -48 \text{ and } 2ab = -14$$

$$\therefore a^2 - b^2 = -48 \text{ and } b = \frac{-7}{a}$$

$$\therefore a^2 - \left(\frac{-7}{a}\right)^2 = -48$$

$$\therefore a^2 - \frac{49}{a^2} = -48$$

$$\therefore a^4 - 49 = -48a^2$$

$$\therefore a^4 + 48a^2 - 49 = 0$$

$$\therefore (a^2 + 49)(a^2 - 1) = 0$$

$$\therefore a^2 = -49 \text{ or } a^2 = 1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -49$$

$$\therefore a^2 = 1$$

$$\therefore a = \pm 1$$

$$\text{When } a = 1, b = -\frac{7}{1} = -7$$

$$\text{When } a = -1, b = \frac{-7}{-1} = 7$$

$$\therefore \sqrt{-48 - 14i} = \pm (1 - 7i)$$

$$\therefore x = \frac{(5 - i) \pm (1 - 7i)}{2}$$

$$\therefore x = \frac{5 - i + 1 - 7i}{2} \text{ or } x = \frac{5 - i - 1 + 7i}{2}$$

$$\therefore x = 3 - 4i \text{ or } x = 2 + 3i.$$

Exercise 3.2 | Q 4.4 | Page 40

Solve the following quadratic equation: $(2 + i)x^2 - (5 - i)x + 2(1 - i) = 0$

SOLUTION

Given equation is

$$(2 + i)x^2 - (5 - i)x + 2(1 - i) = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2 + i, b = -(5 - i), c = 2(1 - i)$$

$$\text{Discriminant} = b^2 - 4ac$$

$$= [-5 - i]^2 - 4 \times (2 + i) \times 2(1 - i)$$

$$= 25 - 10i + i^2 - 8(2 + i)(1 - i)$$

$$= 25 - 10i + i^2 - 8(2 - 2i + i - i^2)$$

$$= 25 - 10i - 1 - 8(2 - i + 1) \quad \dots[\because i^2 = -1]$$

$$= 25 - 10i - 1 - 16 + 8i - 8$$

$$= -2i$$

So, the given equation has complex roots.

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-[-(5 - i)] \pm \sqrt{-2i}}{2(2 + i)}$$

$$= \frac{(5 - i) \pm \sqrt{-2i}}{2(2 + i)}$$

$$\text{Let } \sqrt{-2i} = a + bi, \text{ where } a, b \in \mathbb{R}$$

Squaring on both sides, we get

$$-2i = a^2 + b^2i^2 + 2abi$$

$$\therefore 0 - 2i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 0 \text{ and } 2ab = -2$$

$$\therefore a^2 - b^2 = 0 \text{ and } b = -\frac{1}{a}$$

$$\therefore a^2 - \left(\frac{-1}{a}\right)^2 = 0$$

$$\therefore a^2 - \frac{1}{a^2} = 0$$

$$\therefore a^4 - 1 = 0$$

$$\therefore (a^2 - 1)(a^2 + 1) = 0$$

$$\therefore a^2 = 1 \text{ or } a^2 = -1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -1$$

$$\therefore a^2 = 1$$

$$\therefore a = \pm 1$$

When $a = 1, b = -1$

When $a = -1, b = 1$

$$\therefore \sqrt{-2i} = \pm (1 - i)$$

$$\therefore x = \frac{(5 - i) \pm (1 - i)}{2(2 + i)}$$

$$\therefore x = \frac{5 - i + 1 - i}{2(2 + i)} \text{ or } x = \frac{5 - i - 1 + i}{2(2 + i)}$$

$$\therefore x = \frac{6 - 2i}{2(2 + i)} \text{ or } x = \frac{4}{2(2 + i)}$$

$$\therefore x = \frac{2(3 - i)}{2(2 + i)} \text{ or } x = \frac{2}{2 + i}$$

$$\therefore x = \frac{3 - i}{2 + i} \text{ or } x = \frac{2(2 - i)}{(2 + i)(2 - i)}$$

$$\therefore x = \frac{(3 - i)(2 - i)}{(2 + i)(2 - i)} \text{ or } x = \frac{2(2 - i)}{4 - i^2}$$

$$\therefore x = \frac{6 - 5i + i^2}{4 - i^2} \text{ or } x = \frac{4 - 2i}{4 - i^2}$$

$$\therefore x = \frac{5 - 5i}{5} \text{ or } x = \frac{4 - 2i}{5} \quad \dots[\because i^2 = -1]$$

$$\therefore x = 1 - i \text{ or } x = \frac{4}{5} - \frac{2i}{5}.$$

EXERCISE 3.3 [PAGE 42]

Exercise 3.3 | Q 1.1 | Page 42

If ω is a complex cube root of unity, show that $(2 - \omega)(2 - \omega^2) = 7$

SOLUTION

ω is a complex cube root of unity.

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2 \text{ and } \omega + \omega^2 = -1$$

$$\text{L.H.S.} = (2 - \omega)(2 - \omega^2)$$

$$= 4 - 2\omega^2 - 2\omega + \omega^3$$

$$= 4 - 2(\omega^2 + \omega) + 1$$

$$= 4 - 2(-1) + 1$$

$$= 4 + 2 + 1$$

$$= 7$$

$$= \text{R.H.S.}$$

Exercise 3.3 | Q 1.2 | Page 42

If ω is a complex cube root of unity, show that $(2 + \omega + \omega^2)^3 - (1 - 3\omega + \omega^2)^3 = 65$

SOLUTION

ω is a complex cube root of unity.

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{L.H.S.} = (2 + \omega + \omega^2)^3 - (1 - 3\omega + \omega^2)^3$$

$$= [2 + (\omega + \omega^2)]^3 - [-3\omega + (1 + \omega^2)]^3$$

$$= (2 - 1)^3 - (-3\omega - \omega)^3$$

$$= 1^3 - (-4\omega)^3$$

$$= 1 + 64\omega^3$$

$$= 1 + 64(1)$$

$$= 65$$

$$= \text{R.H.S.}$$

Exercise 3.3 | Q 1.3 | Page 42

If ω is a complex cube root of unity, show that $\frac{(a + b\omega + c\omega^2)}{c + a\omega + b\omega^2} = \omega^2$

SOLUTION

ω is a complex cube root of unity.

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\begin{aligned} \text{L.H.S.} &= \frac{a + b\omega + c\omega^2}{c + a\omega + b\omega^2} \\ &= \frac{a\omega^3 + b\omega^4 + c\omega^2}{c + a\omega + b\omega^2} \quad \dots [\because \omega^3 = 1, \therefore \omega^4 = \omega] \\ &= \frac{\omega^2(c + a\omega + b\omega^2)}{c + a\omega + b\omega^2} \\ &= \omega^2 \\ &= \text{R.H.S.} \end{aligned}$$

Exercise 3.3 | Q 2.1 | Page 42

If ω is a complex cube root of unity, find the value of $\omega + \frac{1}{\omega}$

SOLUTION

ω is a complex cube root of unity

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2 \text{ and } \omega + \omega^2 = -1$$

$$\omega + \frac{1}{\omega} = \frac{\omega^2 + 1}{\omega} = \frac{-\omega}{\omega} = -1.$$

Exercise 3.3 | Q 2.2 | Page 42

If ω is a complex cube root of unity, find the value of $\omega^2 + \omega^3 + \omega^4$

SOLUTION

ω is a complex cube root of unity.

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2 \text{ and } \omega + \omega^2 = -1$$

$$\begin{aligned}
 &\omega^2 + \omega^3 + \omega^4 \\
 &= \omega^2(1 + \omega + \omega^2) \\
 &= \omega^2(0) \\
 &= 0.
 \end{aligned}$$

Exercise 3.3 | Q 2.3 | Page 42

If ω is a complex cube root of unity, find the value of $(1 + \omega^2)^3$

SOLUTION

ω is a complex cube root of unity

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

Also, $1 + \omega^2 = -\omega$, $1 + \omega = -\omega^2$ and $\omega + \omega^2 = -1$

$$(1 + \omega^2)^3 = (-\omega)^3 = -\omega^3 = -1.$$

Exercise 3.3 | Q 2.4 | Page 42

If ω is a complex cube root of unity, find the value of $(1 - \omega - \omega^2)^3 + (1 - \omega + \omega^2)^3$

SOLUTION

ω is a complex cube root of unity

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

Also, $1 + \omega^2 = -\omega$, $1 + \omega = -\omega^2$ and $\omega + \omega^2 = -1$

$$\begin{aligned}
 &(1 - \omega - \omega^2)^3 + (1 - \omega + \omega^2)^3 \\
 &= [1 - (\omega + \omega^2)]^3 + [(1 + \omega^2) - \omega]^3 \\
 &= [1 - (-1)]^3 + (-\omega - \omega)^3 \\
 &= 2^3 + (-2\omega)^3 \\
 &= 8 - 8\omega^3 \\
 &= 8 - 8(1) \\
 &= 0
 \end{aligned}$$

If ω is a complex cube root of unity, find the value of $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$

SOLUTION

ω is a complex cube root of unity

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2 \text{ and } \omega + \omega^2 = -1$$

$$\begin{aligned} & (1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8) \\ &= (1 + \omega)(1 + \omega^2)(1 + \omega)(1 + \omega^2) \quad \dots [\because \omega^3 = 1, \therefore \omega^4 = \omega] \\ &= (-\omega^2)(-\omega)(-\omega^2)(-\omega) \\ &= \omega^6 \\ &= (\omega^3)^2 \\ &= (1)^2 \\ &= 1. \end{aligned}$$

Exercise 3.3 | Q 3 | Page 42

If α and β are the complex cube roots of unity, show that $\alpha^2 + \beta^2 + \alpha\beta = 0$.

SOLUTION

α and β are the complex cube roots of unity.

$$\begin{aligned} \therefore \alpha &= \frac{-1 + i\sqrt{3}}{2} \text{ and } \beta = \frac{-1 - i\sqrt{3}}{2} \\ \therefore \alpha\beta &= \left(\frac{-1 + i\sqrt{3}}{2} \right) \left(\frac{-1 - i\sqrt{3}}{2} \right) \\ &= \frac{(-1)^2 - (i\sqrt{3})^2}{4} \\ &= \frac{1 - (-1)(3)}{4} \quad \dots [\because i^2 = -1] \end{aligned}$$

$$= \frac{1+3}{4}$$

$$\therefore \alpha\beta = 1$$

$$\text{Also, } \alpha + \beta = \frac{-1 + i\sqrt{3}}{2} + \frac{-1 - i\sqrt{3}}{2}$$

$$= \frac{-1 + i\sqrt{3} - 1 - i\sqrt{3}}{2}$$

$$= \frac{-2}{2}$$

$$\therefore \alpha + \beta = -1$$

$$\text{L.H.S.} = \alpha^2 + \beta^2 + \alpha\beta$$

$$= \alpha^2 + 2\alpha\beta + \beta^2 + \alpha\beta - 2\alpha\beta \quad \dots[\text{Adding and subtracting } 2\alpha\beta]$$

$$= (\alpha^2 + 2\alpha\beta + \beta^2) - \alpha\beta$$

$$= (\alpha + \beta)^2 - \alpha\beta$$

$$= (-1)^2 - 1$$

$$= 0$$

$$= \text{R.H.S.}$$

Exercise 3.3 | Q 4 | Page 42

If $x = a + b$, $y = \alpha a + \beta b$ and $z = a\beta + b\alpha$, where α and β are the complex cube roots of unity, show that $xyz = a^3 + b^3$.

SOLUTION

$x = a + b$, $y = \alpha a + \beta b$ and $z = a\beta + b\alpha$
 α and β are the complex cube roots of unity.

$$\therefore \alpha = \frac{-1 + i\sqrt{3}}{2} \quad \text{and} \quad \beta = \frac{-1 - i\sqrt{3}}{2}$$

$$\therefore \alpha\beta = \left(\frac{-1 + i\sqrt{3}}{2} \right) \left(\frac{-1 - i\sqrt{3}}{2} \right)$$

$$\begin{aligned}
&= \frac{(-1)^2 - (i\sqrt{3})^2}{4} \\
&= \frac{1 - (-1)(3)}{4} \quad \dots[\because i^2 = -1] \\
&= \frac{1 + 3}{4}
\end{aligned}$$

$$\therefore \alpha\beta = 1$$

$$\begin{aligned}
\text{Also, } \alpha + \beta &= \frac{-1 + i\sqrt{3}}{2} + \frac{-1 - i\sqrt{3}}{2} \\
&= \frac{-1 + i\sqrt{3} - 1 - i\sqrt{3}}{2} \\
&= \frac{-2}{2}
\end{aligned}$$

$$\therefore \alpha + \beta = 1$$

$$\begin{aligned}
\text{L.H.S.} &= xyz = (a + b)(\alpha a + \beta b)(a\beta + b\alpha) \\
&= (a + b)(\alpha\beta a^2 + \alpha^2 ab + \beta^2 ab + \alpha\beta b^2) \\
&= (a + b)[1 \cdot (a^2) + (\alpha^2 + \beta^2)ab + 1 \cdot (b^2)] \\
&= (a + b) \{a^2 + [(\alpha + \beta)^2 - 2\alpha\beta]ab + b^2\} \\
&= (a + b) \{a^2 + [(-1)^2 - 2(1)]ab + b^2\} \\
&= (a + b) [a^2 + (1 - 2)ab + b^2] \\
&= (a + b)(a^2 - ab + b^2) \\
&= a^3 + b^3 \\
&= \text{R.H.S.}
\end{aligned}$$

Exercise 3.3 | Q 5.1 | Page 42

If ω is a complex cube root of unity, then prove the following: $(\omega^2 + \omega - 1)^3 = -8$

SOLUTION

ω is a complex cube root of unity

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2$$

$$\text{and } \omega + \omega^2 = -1$$

$$\text{L.H.S.} = (\omega^2 + \omega - 1)^3$$

$$= (-1 - 1)^3$$

$$= (-2)^3$$

$$= -8$$

$$= \text{R.H.S.}$$

Exercise 3.3 | Q 5.2 | Page 42

If ω is a complex cube root of unity, then prove the

following: $(a + b) + (a\omega + b\omega^2) + (a\omega^2 + b\omega) = 0.$

SOLUTION

ω is a complex cube root of unity.

$$\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0$$

$$\text{Also, } 1 + \omega^2 = -\omega, 1 + \omega = -\omega^2$$

$$\text{and } \omega + \omega^2 = -1$$

$$\text{L.H.S.} = (a + b) + (a\omega + b\omega^2) + (a\omega^2 + b\omega)$$

$$= (a + a\omega + a\omega^2) + (b + b\omega + b\omega^2)$$

$$= a(1 + \omega + \omega^2) + b(1 + \omega + \omega^2)$$

$$= a(0) + b(0)$$

$$= 0$$

$$= \text{R.H.S.}$$

MISCELLANEOUS EXERCISE 3 [PAGES 42 - 43]**Miscellaneous Exercise 3 | Q 1 | Page 42**

Find the value of $\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}}.$

SOLUTION

$$\begin{aligned}
& \frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} \\
&= \frac{i^{10}(i^{582} + i^{580} + i^{578} + i^{576} + i^{574})}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} \\
&= i^{10} \\
&= (i^4)^2 \cdot i^2 \\
&= (1)^2 (-1) \\
&= -1.
\end{aligned}$$

Miscellaneous Exercise 3 | Q 2 | Page 42

Find the value of $\sqrt{-3} \times \sqrt{-6}$.

SOLUTION

$$\begin{aligned}
\sqrt{-3} \times \sqrt{-6} &= \sqrt{3} \times \sqrt{-1} + \sqrt{6} \times \sqrt{-1} \\
&= \sqrt{3}i \times \sqrt{6}i \\
&= \sqrt{18}i^2 \\
&= -3\sqrt{2} \quad \dots[\because i^2 = -1]
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.01 | Page 42

Simplify the following and express in the form $a + ib$: $3 + \sqrt{-64}$

SOLUTION

$$\begin{aligned}
& 3 + \sqrt{-64} \\
&= 3 + \sqrt{64} \cdot \sqrt{-1} \\
&= 3 + 8i
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.02 | Page 42

Simplify the following and express in the form $a + ib$: $(2i^3)^2$

SOLUTION

$$\begin{aligned}
(2i^3)^2 &= 4i^6 \\
&= 4(i^2)^3 \\
&= 4(-1)^3 \quad \dots[\because i^2 = -1] \\
&= -4 \\
&= -4 + 0i
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.03 | Page 42

Simplify the following and express in the form $a + ib$: $(2 + 3i)(1 - 4i)$

SOLUTION

$$\begin{aligned}
(2 + 3i)(1 - 4i) \\
&= 2 - 8i + 3i - 12i^2 \\
&= 2 - 5i - 12(-1) \quad \dots[\because i^2 = -1] \\
&= 14 - 5i
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.04 | Page 42

Simplify the following and express in the form $a + ib$: $\frac{5}{2}i(-4 - 3i)$

SOLUTION

$$\begin{aligned}
&\frac{5}{2}i(-4 - 3i) \\
&= \frac{5}{2}i(-4i - 3i^2) \\
&= \frac{5}{2}[-4i - 3(-1)] \quad \dots[\because i^2 = -1] \\
&= \frac{5}{2}(3 - 4i) \\
&= \frac{15}{2} - 10i
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.05 | Page 42

Simplify the following and express in the form $a + ib$: $(1 + 3i)^2 (3 + i)$

SOLUTION

$$\begin{aligned}
(1 + 3i)^2 (3 + i) \\
&= (1 + 6i + 9i^2)(3 + i) \\
&= (1 + 6i - 9)(3 + i) \quad \dots[\because i^2 = -1]
\end{aligned}$$

$$\begin{aligned}
&= (-8 + 6i)(3 + i) \\
&= -24 - 8i + 18i + 6i^2 \\
&= -24 + 10i + 6(-1) \\
&= -24 + 10i - 6 \\
&= -30 + 10i
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.06 | Page 42

Simplify the following and express in the form $a + ib$: $\frac{4 + 3i}{1 - i}$

SOLUTION

$$\begin{aligned}
\frac{4 + 3i}{1 - i} &= \frac{(4 + 3i)(1 + i)}{(1 - i)(1 + i)} \\
&= \frac{4 + 4i + 3i + 3i^2}{1 - i^2} \\
&= \frac{4 + 7i + 3(-1)}{1 - (-1)} \quad \dots [\because i^2 = -1] \\
&= \frac{1 + 7i}{2} \\
&= \frac{1}{2} + \frac{7}{2}i.
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.07 | Page 42

Simplify the following and express in the form $a + ib$: $\left(1 + \frac{2}{i}\right)\left(3 + \frac{4}{i}\right)(5 + i)^{-1}$

SOLUTION

$$\begin{aligned}& \left(1 + \frac{2}{i}\right) \left(3 + \frac{4}{i}\right) (5 + i)^{-1} \\&= \frac{(i + 2)}{i} \cdot \frac{(3i + 4)}{i} \cdot \frac{1}{5 + i} \\&= \frac{3i^2 + 4i + 6i + 8}{i^2(5 + i)} \\&= \frac{-3 + 10i + 8}{-1(5 + i)} \quad \dots[\because i^2 = -1] \\&= \frac{(5 + 10i)}{-(5 + i)} \\&= \frac{(5 + 10i)(5 - i)}{-(4 + i)(5 - i)} \\&= \frac{25 - 5i + 50i - 10i^2}{-(25 - i^2)} \\&= \frac{25 + 45i - 10(-1)}{-[25 - (-1)]} \\&= \frac{35 + 45i}{-26} \\&= \frac{-35}{26} - \frac{45}{26}i\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.08 | Page 42

Simplify the following and express in the form $a + ib$: $\frac{\sqrt{5} + \sqrt{3}i}{\sqrt{5} - \sqrt{3}i}$

SOLUTION

$$\begin{aligned}
& \frac{\sqrt{5} + \sqrt{3}i}{\sqrt{5} - \sqrt{3}i} \\
&= \frac{(\sqrt{5} + \sqrt{3}i)(\sqrt{5} + \sqrt{3}i)}{(\sqrt{5} - \sqrt{3}i)(\sqrt{5} + \sqrt{3}i)} \\
&= \frac{5 + 2\sqrt{15}i + 3i^2}{5 - 3i^2} \\
&= \frac{5 + 2\sqrt{15}i + 3(-1)}{5 - 3(-1)} \quad \dots[\because i^2 = -1] \\
&= \frac{2 + 2\sqrt{15}i}{8} \\
&= \frac{1 + \sqrt{15}i}{4} \\
&= \frac{1}{4} + \frac{\sqrt{15}i}{4}.
\end{aligned}$$

Miscellaneous Exercise 3 | Q 3.09 | Page 43

Simplify the following and express in the form $a + ib$: $\frac{3i^5 + 2i^7 + i^9}{i^6 + 2i^8 + 3i^{18}}$

SOLUTION

$$\begin{aligned}
& \frac{3i^5 + 2i^7 + i^9}{i^6 + 2i^8 + 3i^{18}} \\
&= \frac{3(i^4 \cdot i) + 2(i^4 \cdot i^3) + (i^4)^2 \cdot i}{i^4 \cdot i^2 + 2(i^4) + 3(i^2)^9} \\
&= \frac{3(1) \cdot i + 2(1)(-i) + (1)^2 \cdot i}{(1)(-1) + 2(1)^2 + 3(-1)^9} \quad \dots[\because i^2 = -1, i^3 = -i, i^4 = 1]
\end{aligned}$$

$$= \frac{3i - 2i + i}{-1 + 2 - 3}$$

$$= \frac{2i}{-2}$$

$$= -i$$

$$= 0 - i$$

Miscellaneous Exercise 3 | Q 3.1 | Page 43

Simplify the following and express in the form $a + ib$: $\frac{5 + 7i}{4 + 3i} + \frac{5 + 7i}{4 - 3i}$

SOLUTION

$$\frac{5 + 7i}{4 + 3i} + \frac{5 + 7i}{4 - 3i}$$

$$= (5 + 7i) \left[\frac{1}{4 + 3i} + \frac{1}{4 - 3i} \right]$$

$$= (5 + 7i) \left[\frac{4 - 3i + 4 + 3i}{(4 + 3i)(4 - 3i)} \right]$$

$$= (5 + 7i) \left[\frac{8}{16 - 9i^2} \right]$$

$$= (5 + 7i) \left[\frac{8}{16 - 9(-1)} \right] \quad \dots [\because i^2 = -1]$$

$$= \frac{8(5 + 7i)}{25}$$

$$= \frac{40 + 56i}{25}$$

$$= \frac{40}{25} + \frac{56}{25}i$$

$$= \frac{8}{5} + \frac{56}{25}i.$$

Miscellaneous Exercise 3 | Q 4.1 | Page 43

Solve the following equation for $x, y \in \mathbb{R}$: $(4 - 5i)x + (2 + 3i)y = 10 - 7i$

SOLUTION

$$(4 - 5i)x + (2 + 3i)y = 10 - 7i$$

$$\therefore (4x + 2y) + (3y - 5x)i = 10 - 7i$$

Equating real and imaginary parts, we get

$$4x + 2y = 10$$

$$\text{i.e., } 2x + y = 5 \quad \dots(i)$$

$$\text{and } 3y - 5x = -7 \quad \dots(ii)$$

Equation (i) $\times$ 3 – equation (ii) gives

$$11x = 22$$

$$\therefore x = 2$$

Putting $x = 2$ in (i), we get

$$2(2) + y = 5$$

$$\therefore y = 1$$

$$\therefore x = 2 \text{ and } y = 1.$$

Miscellaneous Exercise 3 | Q 4.2 | Page 43

Solve the following equation for $x, y \in \mathbb{R}$: $(1 - 3i)x + (2 + 5i)y = 7 + i$

SOLUTION

$$(1 - 3i)x + (2 + 5i)y = 7 + i$$

$$\therefore (x + 2y) + (-3x + 5y)i = 7 + i$$

Equating real and imaginary parts, we get

$$x + 2y = 7 \quad \dots(i)$$

$$\text{and } -3x + 5y = 1 \quad \dots(ii)$$

Equation (i) $\times$ 3 + equation (ii) gives

$$11y = 22$$

$$\therefore y = 2$$

Putting $y = 2$ in (i), we get

$$x + 2(2) = 7$$

$$\therefore x = 3$$

$$\therefore x = 3 \text{ and } y = 2.$$

Miscellaneous Exercise 3 | Q 4.3 | Page 43

Solve the following equation for $x, y \in \mathbb{R}$: $\frac{x + iy}{2 + 3i} = 7 - i$

SOLUTION

$$\frac{x + iy}{2 + 3i} = 7 - i$$

$$\therefore x + iy = (7 - i)(2 + 3i)$$

$$\therefore x + iy = 14 + 21i - 2i - 3i^2$$

$$\therefore x + iy = 14 + 19i - 3(-1) \quad \dots[\because i^2 = -1]$$

$$\therefore x + iy = 17 + 19i$$

Equating real and imaginary parts, we get

$$x = 17 \text{ and } y = 19$$

Miscellaneous Exercise 3 | Q 4.4 | Page 43

Solve the following equation for $x, y \in \mathbb{R}$: $(x + iy)(5 + 6i) = 2 + 3i$

SOLUTION

$$(x + iy)(5 + 6i) = 2 + 3i$$

$$\therefore x + iy = \frac{2 + 3i}{5 + 6i}$$

$$\therefore x + iy = \frac{(2 + 3i)(5 - 6i)}{(5 + 6i)(5 - 6i)}$$

$$= \frac{10 - 12i + 15i - 18i^2}{25 - 36i^2} \quad \dots[\because i^2 = -1]$$

$$= \frac{10 + 3i - 18(-1)}{25 - 36(-1)}$$

$$\therefore x + iy = \frac{28 + 3i}{61}$$

$$= \frac{28}{61} + \frac{3}{61}i$$

Equating real and imaginary parts, we get

$$x = \frac{28}{61} \text{ and } y = \frac{3}{61}.$$

Miscellaneous Exercise 3 | Q 4.5 | Page 43

Solve the following equation for $x, y \in \mathbb{R}$: $2x + i^9 y (2 + i) = x i^7 + 10 i^{16}$

SOLUTION

$$2x + i^9 y (2 + i) = x i^7 + 10 i^{16}$$

$$\therefore 2x + (i^4)^2 \cdot i \cdot y (2 + i) = x (i^2)^3 \cdot i + 10 \cdot (i^4)^4$$

$$\therefore 2x + (1)^2 \cdot i y (2 + i) = x (-1)^3 \cdot i + 10 (1)^4 \quad \dots [\because i^2 = -1, i^4 = 1]$$

$$\therefore 2x + 2yi + yi^2 = -xi + 10$$

$$\therefore 2x + 2yi - y + xi = 10$$

$$\therefore (2x - y) + (x + 2y)i = 10 + 0 \cdot i$$

Equating real and imaginary parts, we get

$$2x - y = 10 \quad \dots(i)$$

$$\text{and } x + 2y = 0 \quad \dots(ii)$$

Equation (i) $\times$ 2 + equation (ii) gives

$$5x = 20$$

$$\therefore x = 4$$

Putting $x = 4$ in (i), we get

$$2(4) - y = 10$$

$$\therefore y = 8 - 10$$

$$\therefore y = -2$$

$$\therefore x = 4 \text{ and } y = -2$$

Miscellaneous Exercise 3 | Q 5.1 | Page 43

Find the value of : $x^3 + 2x^2 - 3x + 21$, if $x = 1 + 2i$

SOLUTION

$$x = 1 + 2i$$

$$\therefore x - 1 = 2i$$

$$\therefore (x - 1)^2 = 4i^2$$

$$\therefore x^2 - 2x + 1 = -4 \quad \dots [\because i^2 = -1]$$

$$\therefore x^2 - 2x + 5 = 0 \quad \dots(i)$$

$$\begin{array}{r}
 \div x^2 - 2x + 5 \overline{) x^3 + 2x^2 - 3x + 21} \\
 \underline{x^3 - 2x^2 + 5x} \\
 4x^2 - 8x + 21 \\
 \underline{4x^2 - 8x + 20} \\
 1
 \end{array}$$

$$\begin{aligned}
 &\therefore x^3 + 2x^2 - 3x + 21 \\
 &= (x^2 - 2x + 5)(x + 4) + 1 \\
 &= 0(x + 4) + 1 \quad \dots[\text{From (i)}] \\
 &= 0 + 1 \\
 &\therefore x^3 + 2x^2 - 3x + 21 = 1
 \end{aligned}$$

Miscellaneous Exercise 3 | Q 5.2 | Page 43

Find the value of : $x^3 - 5x^2 + 4x + 8$, if $x = \frac{10}{3 - i}$

SOLUTION

$$\begin{aligned}
 x &= \frac{10}{3 - i} \\
 \therefore x &= \frac{10(3 + i)}{(3 - i)(3 + i)} \\
 &= \frac{10(3 + i)}{9 - i^2}
 \end{aligned}$$

$$= \frac{10(3 + i)}{9 - (-1)} \quad \dots [\because i^2 = -1]$$

$$= \frac{10(3 + i)}{10}$$

$$\therefore x = 3 + i$$

$$\therefore x - 3 = i$$

$$\therefore (x - 3)^2 = i^2$$

$$\therefore x^2 - 6x + 9 = -1 \quad \dots [\because i^2 = -1]$$

$$\therefore x^2 - 6x + 10 = 0 \quad \dots(i)$$

$$\begin{array}{r} x+1 \\ x^2-6x+10 \overline{) x^3+5x^2+4x+8} \\ \underline{x^3-6x^2+10x} \\ 11x^2-6x \\ \underline{11x^2-66x+110} \\ 60x-102 \\ \underline{60x-60} \\ -42 \end{array}$$

$$\therefore x^3 - 5x^2 + 4x + 8$$

$$= (x^2 - 6x + 10)(x + 1) - 2$$

$$= 0. (x + 1) - 2 \quad \dots[\text{From (i)}]$$

$$= 0 - 2$$

$$\therefore x^3 - 5x^2 + 4x + 8 = -2.$$

Miscellaneous Exercise 3 | Q 5.3 | Page 43

Find the value of: $x^3 - 3x^2 + 19x - 20$, if $x = 1 - 4i$

SOLUTION

$$x = 1 - 4i$$

$$\therefore x - 1 = -4i$$

$$\begin{aligned}\therefore (x-1)^2 &= 16i^2 \\ \therefore x^2 - 2x + 1 &= -16 \quad \dots[\because i^2 = -1] \\ \therefore x^2 - 2x + 17 &= 0 \quad \dots(i)\end{aligned}$$

$$\begin{array}{r} x-1 \\ x^2-2x+17 \overline{) x^3-3x^2+19x^2-20} \\ \underline{x^3-2x^2+17x} \\ -x^2+2x-20 \\ \underline{-x^2+2x-17x} \\ -3 \end{array}$$

$$\begin{aligned}\therefore x^3 - 3x^2 + 19x - 20 &= (x^2 - 2x + 17)(x - 1) - 3 \\ &= 0 \cdot (x - 1) - 3 \quad \dots[\text{From (i)}] \\ &= 0 - 3 \\ \therefore x^3 - 3x^2 + 19x - 20 &= -3\end{aligned}$$

Miscellaneous Exercise 3 | Q 6.1 | Page 43

Find the square root of: $-16 + 30i$

SOLUTION

Let $\sqrt{-16 + 30i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$-16 + 30i = a^2 + b^2i^2 + 2abi$$

$$\therefore -16 + 30i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = -16 \text{ and } 2ab = 30$$

$$\therefore a^2 - b^2 = -16 \text{ and } b = \frac{15}{a}$$

$$\therefore a^2 - \frac{225}{a^2} = -16$$

$$\therefore a^4 - 225 - 16a^2$$

$$\therefore a^4 + 16a^2 - 225 = 0$$

$$\therefore (a^2 + 25)(a^2 - 9) = 0$$

$$\therefore a^2 = -25 \text{ or } a^2 = 9$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -25$$

$$\therefore a^2 = 9$$

$$\therefore a = \pm 3$$

$$\text{When } a = 3, b = \frac{15}{3} = 5$$

$$\text{When } a = -3, b = \frac{15}{-3} = -5$$

$$\therefore \sqrt{-16 + 30i} = \pm (3 + 5i)$$

Miscellaneous Exercise 3 | Q 6.2 | Page 43

Find the square root of $15 - 8i$

SOLUTION

Let $\sqrt{15 - 8i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$15 - 8i = a^2 + b^2i^2 + 2abi$$

$$\therefore 15 - 8i = (a^2 - b^2) + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 15 \text{ and } 2ab = -8$$

$$\therefore a^2 - b^2 = 15 \text{ and } b = \frac{-4}{a}$$

$$\therefore a^2 \left(\frac{-4}{a} \right)^2 = 15$$

$$\therefore a^2 - \frac{16}{a^2} = 15$$

$$\therefore a^4 - 16 = 15a^2$$

$$\therefore a^4 - 15a^2 - 16 = 0$$

$$\therefore (a^2 - 16)(a^2 + 1) = 0$$

$$\therefore a^2 = 16 \text{ or } a^2 = -1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -1$$

$$\therefore a^2 = 16$$

$$\therefore a = \pm 4$$

$$\text{When } a = 4, b = \frac{-4}{4} = -1$$

$$\therefore \sqrt{15 - 8i} = \pm (4 - i).$$

Miscellaneous Exercise 3 | Q 6.3 | Page 43

Find the square root of: $2 + 2\sqrt{3}i$

SOLUTION

Let $\sqrt{2 + 2\sqrt{3}i} = a + bi$, where $a, b \in \mathbb{R}$.

Squaring on both sides, we get

$$2 + 2\sqrt{3}i = a^2 + b^2i^2 + 2abi$$

$$\therefore 2 + 2\sqrt{3}i = a^2 - b^2 + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 2 \text{ and } 2ab = 2\sqrt{3}$$

$$\therefore a^2 - b^2 = 2 \text{ and } b = \frac{\sqrt{3}}{a}$$

$$\therefore a^2 - \left(\frac{\sqrt{3}}{a}\right)^2 = 2$$

$$\therefore a^2 - \frac{3}{a^2} = 2$$

$$\therefore a^4 - 3 = 2a^2$$

$$\therefore a^4 - 2a^2 - 3 = 0$$

$$\therefore (a^2 - 3)(a^2 + 1) = 0$$

$$\therefore a^2 = 3 \text{ or } a^2 = -1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -1$$

$$\therefore a^2 = 3$$

$$\therefore a = \pm \sqrt{3}$$

$$\text{When } a = \sqrt{3}, b = \frac{\sqrt{3}}{\sqrt{3}} = 1$$

$$\text{When } a = -\sqrt{3}, b = \frac{\sqrt{3}}{-\sqrt{3}} = -1$$

$$\therefore \sqrt{2 + 2\sqrt{3}i} = \pm(\sqrt{3} + i).$$

Miscellaneous Exercise 3 | Q 6.4 | Page 43

Find the square root of : $18i$

SOLUTION

Let $\sqrt{18i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$18i = a^2 + b^2i^2 + 2abi$$

$$\therefore 0 + 18i = a^2 - b^2 + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 0 \text{ and } 2ab = 18$$

$$\therefore a^2 - b^2 = 0 \text{ and } b = \frac{9}{a}$$

$$\therefore a^2 - \left(\frac{9}{a}\right)^2 = 0$$

$$\therefore a^2 - \frac{81}{a^2} = 0$$

$$\therefore a^4 - 81 = 0$$

$$\therefore (a^2 - 9)(a^2 + 9) = 0$$

$$\therefore a^2 = 9 \text{ or } a^2 = -9$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -9$$

$$\therefore a^2 = 9$$

$$\therefore a = \pm 3$$

$$\text{When } a = 3, b = \frac{9}{3} = 3$$

$$\text{When } a = -3, b = \frac{9}{-3} = -3$$

$$\therefore \sqrt{18i} = \pm(3 + 3i) = \pm 3(1 + i).$$

Miscellaneous Exercise 3 | Q 6.5 | Page 43

Find the square root of: $3 - 4i$

SOLUTION

Let $\sqrt{3 - 4i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$3 - 4i = a^2 + b^2i^2 + 2abi$$

$$\therefore 3 - 4i = a^2 - b^2 + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 3 \text{ and } 2ab = -4$$

$$\therefore a^2 - b^2 = 3 \text{ and } b = \frac{-2}{a}$$

$$\therefore a^2 - \left(-\frac{2}{a}\right)^2 = 3$$

$$\therefore a^2 - \frac{4}{a^2} = 3$$

$$\therefore a^4 - 4 = 3a^2$$

$$\therefore a^4 - 3a^2 - 4 = 0$$

$$\therefore (a^2 - 4)(a^2 + 1) = 0$$

$$\therefore a^2 = 4 \text{ or } a^2 = -1$$

But $a \in \mathbb{R}$

$$\therefore a^2 \neq -1$$

$$\therefore a^2 = 4$$

$$\therefore a = \pm 2$$

$$\text{When } a = 2, b = \frac{-2}{2} = -1$$

$$\text{When } a = -2, b = \frac{-2}{-2} = 1$$

$$\therefore \sqrt{3 - 4i} = \pm (2 - i).$$

Find the square root of: $6 + 8i$

SOLUTION

Let $\sqrt{6 + 8i} = a + bi$, where $a, b \in \mathbb{R}$

Squaring on both sides, we get

$$6 + 8i = a^2 + b^2i^2 + 2abi$$

$$\therefore 6 + 8i = a^2 - b^2 + 2abi \quad \dots[\because i^2 = -1]$$

Equating real and imaginary parts, we get

$$a^2 - b^2 = 6 \text{ and } 2ab = 8$$

$$\therefore a^2 - b^2 = 6 \text{ and } b = \frac{4}{a}$$

$$\therefore a^2 - \left(\frac{4}{a}\right)^2 = 6$$

$$\therefore a^2 - \frac{16}{a^2} = 6$$

$$\therefore a^4 - 16 = 6a^2$$

$$\therefore a^4 - 6a^2 - 16 = 0$$

$$\therefore (a^2 - 8)(a^2 + 2) = 0$$

$$\therefore a^2 = 8 \text{ or } a^2 = -2$$

But

$$\therefore a^2 \neq -2$$

$$\therefore a^2 = 8$$

$$\therefore a = \pm 2\sqrt{2}$$

$$\text{When } a = 2\sqrt{2}, \quad b = \frac{4}{2\sqrt{2}} = \sqrt{2}$$

$$\text{When } a = -2\sqrt{2}, \quad b = \frac{4}{-2\sqrt{2}} = -\sqrt{2}$$

$$\therefore \sqrt{6+8i} = \pm(2\sqrt{2} + \sqrt{2}i) = \pm\sqrt{2}(2+i).$$